

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Indicative content amount of water vapour decreased as Earth cooled, water vapour condensed and oceans formed amount of carbon dioxide decreased evolution of green plants, photosynthesis, CO₂ taken in by plants / algae evolution of marine animals / CO₂ locked in limestone / chalk / carbonates rock remains of marine organisms / (land) plants locked into fossil fuels amount of oxygen increased evolution of green plants, photosynthesis, O₂ released by plants 5-6 marks Good explanation of changes for all three gases <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Basic explanation of changes referring to photosynthesis and condensation of water vapour <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Simple description of changes in percentage of gases <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		No significant impact on the overall level of carbon dioxide ☒						
				A significant increase in the level of carbon dioxide ☐			1	1		
				A significant decrease in the level of carbon dioxide ☐						
		(ii)		Mean atmospheric temperature decreases ☒						
				Mean atmospheric temperature increases ☐			1	1		
				No effect on the mean atmospheric temperature ☐						
		(iii)		Solar radiation decreases because it is reflected by sulfur dioxide ☒						
				Solar radiation increases because it is absorbed by carbon dioxide ☐						
				Solar radiation increases because it is absorbed by carbon dioxide and sulfur dioxide ☐			1	1		
				Solar radiation decreases because it reacts with sulfur dioxide forming sulfuric acid ☐						
				Question 6 total	6	0	3	9	0	0